NARASARAOPETA ENGINEERING COLLEGE (AUTONOMOUS)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

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Batch Number	AG1
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Guide	Dr.S.N.Tirumala Rao M.Tech,Ph.D
Title	Xception Spiking Fractional Neural Network for Oral Squamous Cell Carcinoma Classification Based on Histopathological Image
Domain/Technology	DEEP LEARNING
Base Paper Link	https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=11016737
Dataset Link	https://data.mendeley.com/datasets/ftmp4cvtmb/2
Software Requirements	Browser: Any latest browser like Chrome Operating System: Windows 7 Server or later Python (COLAB)
Hardware Requirements	SystemType: Intel Core i5 or above RAM: 8 GB Number of cores:5 Number of Threads: 4
Abstract	Oral Squamous Cell Carcinoma (OSCC) is the leading cancer caused due to the immense growth of malignant cells in different regions of the oral cavity particularly in the area of the neck and head. The habits of smoking, betel nuts, and chewing tobacco that mainly affect the pharynx, nasopharynx, viral infection, and oral cavity regions are the major causes of OSCC. OSCC has to be detected at the initial stages to provide accurate treatment and to assess the severity. Thus, various screening and detection models are used to accurately detect OSCC with great significance for reducing morbidity and mortality. In this research, a novel deep learning model Xception Spiking Fractional Neural Network (XSFN-Net) is introduced to classify OSCC from the histopathology images. Here, the histopathology images are enhanced initially using the Medav filter. The pixel-wise image segmentation is performed using Parallel Reverse Attention Network along with the designed loss function from the enhanced image. To improve the accuracy of the proposed **XSFN-Net model** for oral squamous cell carcinoma (OSCC) classification, several strategies can be applied across the data, model, and training stages.To increase the size of the dataset.

Signature of the student(s) **Signature of the Guide** **Signature of the project coordinator**